The Common Support Interval of Coordinate Factorization and Term Presentation in Finite Boolean Functions

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Abstract

Let

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

be a finite typed Boolean function with determinational support R_f. We consider two families of coordinate sets associated with f: the factorization-support family $\mathfrak{F}_N(f)$, consisting of those $M \subseteq N$ through which f factors, and the presentation-support family $\mathfrak{S}_N(f)$, consisting of the exact syntactic supports of term presentations of f in a fixed Boolean term language. We prove that

$$\mathfrak{F}_N(f) = \mathfrak{S}_N(f) = [R_f, N]_{\subseteq}.$$

Thus a coordinate set supports a factorization of f if and only if it occurs as the exact syntactic support of some presentation of the same typed function on the same frame. In particular, R_f is the common least element of both families. The equality concerns support families only: coordinate factorization and term presentation remain structurally distinct constructions.

Keywords: finite Boolean functions; determinational support; coordinate factorization; syntactic support; term presentation; support interval.
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1 Introduction

Let

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

be a finite typed Boolean function with determinational support

$$R_f \subseteq N.$$

Two different questions arise naturally once the ambient frame N is kept fixed.

The first concerns coordinate factorization: through which subsets

$$M \subseteq N$$

can the function f factor? Volume I [1] establishes the canonical factorization

$$f = f_{R_f} \circ \pi^N_{R_f}$$

and shows that R_f is the least coordinate support through which such a factorization is possible. This determines the lower boundary of the factorization problem, but does not by itself collect all coordinate supports through which f may factor.

The second question concerns presentation. Fix the Boolean term language $\mathcal{L}$ used in III-A. For an $\mathcal{L}$-term τ representing the same typed function f on the same frame N, let S_τ denote its syntactic support. III-A [2] studies the corresponding presentation-support family

$$\mathfrak{S}_N(f)$$

and establishes the presentation-support interval

$$\mathfrak{S}_N(f) = [R_f, N]_\subseteq.$$

Thus every coordinate set between the determinational support and the formal frame occurs as the exact syntactic support of some presentation of f.

These two results raise a direct comparison problem. If R_f is the least support through which f can factor, what is the full family of factorization supports, and how does that family relate to the family of presentation supports?

The present paper answers this question by introducing the factorization-support family

$$\mathfrak{F}_N(f)$$

and proving

$$\mathfrak{F}_N(f) = [R_f, N]_\subseteq.$$

Together with the presentation-support theorem from III-A, this yields the central result

$$\boxed{\mathfrak{F}_N(f) = \mathfrak{S}_N(f) = [R_f, N]_\subseteq.}$$

The equality is an equality of support families. It does not identify coordinate factorization with syntactic presentation. The two constructions retain different structural roles, while admitting exactly the same supports inside the fixed frame N.

The paper proceeds accordingly. We first characterize the factorization-support family, then recall the corresponding presentation-support result, and finally establish their coincidence. The concluding section records the common lower boundary R_f and indicates the remaining question of distinct realizations over a fixed support.

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2 Factorization Supports

Let

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

be a finite typed Boolean function, and let

$$R_f \subseteq N$$

denote its determinational support.

The canonical reduction established in Volume I [1] gives a unique function

$$f_{R_f} : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}$$

such that

$$f = f_{R_f} \circ \pi^N_{R_f}.$$

Moreover, R_f is the least coordinate support through which f can factor: if $M \subseteq N$ and there exists a function

$$g : \mathbf{2}^M \longrightarrow \mathbf{2}$$

such that

$$f = g \circ \pi^N_M,$$

then

$$R_f \subseteq M.$$

These two facts determine the lower boundary of coordinate factorization. We now collect all coordinate supports through which the fixed typed function f can factor.

Definition 2.1 (Factorization-support family)**.** For

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

define

$$\mathfrak{F}_N(f) := \left\{ M \subseteq N : \exists\, g_M : \mathbf{2}^M \longrightarrow \mathbf{2} \text{ such that } f = g_M \circ \pi^N_M \right\}.$$

Thus $\mathfrak{F}_N(f)$ records the coordinate sets that are sufficient to carry a factorization of f, while the ambient frame N remains fixed.

Theorem 2.2 (Factorization-support interval)**.** *For every finite typed Boolean function*

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

one has

$$\boxed{\mathfrak{F}_N(f) = [R_f, N]_{\subseteq} = \{M \subseteq N : R_f \subseteq M\}.}$$

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Proof. Let

$$M \in \mathfrak{F}_N(f).$$

By definition, there exists

$$g_M : \mathbf{2}^M \longrightarrow \mathbf{2}$$

such that

$$f = g_M \circ \pi_M^N.$$

Since R_f is the least coordinate support through which f factors, it follows that

$$R_f \subseteq M.$$

Hence

$$\mathfrak{F}_N(f) \subseteq [R_f, N]_{\subseteq}.$$

Conversely, let

$$R_f \subseteq M \subseteq N.$$

The corresponding coordinate projections satisfy

$$\pi_{R_f}^N = \pi_{R_f}^M \circ \pi_M^N.$$

Using the canonical factorization

$$f = f_{R_f} \circ \pi_{R_f}^N,$$

we obtain

$$f = f_{R_f} \circ \pi_{R_f}^M \circ \pi_M^N.$$

Define

$$f_M := f_{R_f} \circ \pi_{R_f}^M : \mathbf{2}^M \longrightarrow \mathbf{2}.$$

Then

$$f = f_M \circ \pi_M^N,$$

and therefore

$$M \in \mathfrak{F}_N(f).$$

Thus

$$[R_f, N]_{\subseteq} \subseteq \mathfrak{F}_N(f).$$

Consequently,

$$\mathfrak{F}_N(f) = [R_f, N]_{\subseteq}.$$

□

The theorem extends without exception to the case

$$R_f = \varnothing,$$

with the standard convention that $\mathbf{2}^{\varnothing}$ is a singleton. In that case f is constant, and every subset of N is a factorization support.

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3 Presentation Supports

We now consider the same fixed typed Boolean function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

from the side of its term presentations.

Let $\mathcal{L}$ denote the fixed Boolean term language used in III-A [2]. For an $\mathcal{L}$-term τ, let

$$S_\tau \subseteq N$$

denote its syntactic support, that is, the set of coordinates occurring literally in τ.

Define the set of term presentations of f on the fixed frame N by

$$\mathrm{Rep}_N(f) = \left\{\tau : [\![\tau]\!]_N^{\mathcal{L}} = f\right\},$$

and define the corresponding presentation-support family by

$$\mathfrak{S}_N(f) = \{S_\tau : \tau \in \mathrm{Rep}_N(f)\}.$$

As in III-A, the dependence of this family on the fixed language $\mathcal{L}$ is suppressed in the notation.

For every

$$\tau \in \mathrm{Rep}_N(f),$$

one has

$$R_f \subseteq S_\tau \subseteq N.$$

III-A [2] establishes the converse realization statement as well: every coordinate set between R_f and N occurs as the exact syntactic support of some term representing the same typed function f on the same frame N. Consequently,

$$\boxed{\mathfrak{S}_N(f) = [R_f, N]_\subseteq = \{S \subseteq N : R_f \subseteq S\}.}$$

4 The Bridge Theorem

The preceding sections characterize the factorization-support and presentation-support families of the same fixed typed Boolean function by the same interval. This yields the central identification.

Theorem 4.1 (Bridge Theorem)**.** *For every finite typed Boolean function*

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

one has

$$\boxed{\mathfrak{F}_N(f) = \mathfrak{S}_N(f) = [R_f, N]_\subseteq.}$$

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Proof. By the factorization-support interval theorem,

$$\mathfrak{F}_N(f) = [R_f, N]_{\subseteq}.$$

By the presentation-support interval theorem,

$$\mathfrak{S}_N(f) = [R_f, N]_{\subseteq}.$$

Therefore

$$\mathfrak{F}_N(f) = \mathfrak{S}_N(f) = [R_f, N]_{\subseteq}.$$

□

Thus a coordinate set

$$M \subseteq N$$

supports a factorization of f if and only if it occurs as the exact syntactic support of some term presentation of the same typed function on the same frame N.

The equality concerns the families of support sets. It does not identify factorization with presentation: the two constructions have different structural roles, even though they determine the same interval of admissible supports.

5 Conclusion and Further Direction

The Bridge Theorem shows that, for a fixed typed Boolean function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

the coordinate supports admissible for factorization and the syntactic supports realizable by term presentations form exactly the same family:

$$\mathfrak{F}_N(f) = \mathfrak{S}_N(f) = [R_f, N]_{\subseteq}.$$

In particular, the determinational support R_f is the common least element of both families:

$$\boxed{R_f = \min \mathfrak{F}_N(f) = \min \mathfrak{S}_N(f).}$$

Equivalently,

$$R_f = \bigcap_{M \in \mathfrak{F}_N(f)} M = \bigcap_{\tau \in \mathrm{Rep}_N(f)} S_\tau.$$

Within the fixed frame N and the fixed Boolean term language $\mathcal{L}$, the Bridge Theorem therefore closes the question of which supports can occur on the two sides. A further question concerns the relation between the factorization associated with a fixed support and the term presentations carried by that same support

$$M \in [R_f, N]_{\subseteq}.$$

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On the factorization side, such an M supports a factor

$$f_M : \mathbf{2}^M \longrightarrow \mathbf{2}$$

with

$$f = f_M \circ \pi_M^N.$$

On the presentation side, the same support may carry multiple term presentations. One may record them by

$$\mathrm{Rep}_M^N(f) := \{\tau \in \mathrm{Rep}_N(f) : S_\tau = M\}.$$

The relation between these two forms of realization over a common support is left for subsequent study.

References

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